

Notes: Barberis, Huang, and Santos (QJE, 2001)

Prospect Theory and Asset Prices

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1 Big picture

- **Question.** Can a behavioral model based on prospect theory explain the aggregate stock market facts that are difficult for standard consumption-based asset pricing models?
- **Target facts.** The paper aims to explain the high average excess return of stocks, high stock return volatility, time-varying expected returns, return predictability from price-dividend ratios, and the low correlation between stock returns and consumption growth.
- **Core departure from standard theory.** The representative investor obtains utility not only from consumption but also directly from fluctuations in the value of financial wealth.
- **Prospect-theory feature 1: loss aversion.** The investor is more sensitive to reductions in financial wealth than to increases of the same size.
- **Prospect-theory feature 2: prior outcomes.** Loss aversion depends on recent investment performance. Prior gains make subsequent losses less painful; prior losses make subsequent losses more painful.
- **Economic mechanism.** After stock price run-ups, investors become less risk averse because past gains cushion future losses. After price declines, they become more loss averse. This generates endogenous time variation in risk aversion.
- **Pricing mechanism.** A high current price-dividend ratio indicates past gains and hence low risk aversion. Low risk aversion lowers future expected returns. This creates return predictability and mean reversion.
- **Main quantitative result.** In the richer economy where consumption and dividends are imperfectly correlated, the model generates a sizeable equity premium, high stock return volatility, low stock-return/consumption-growth correlation, and predictive variation in expected returns.
- **Key limitation.** A pure loss-aversion model without prior-outcome dependence has trouble generating enough return volatility and predictability. Prior outcomes are essential for the model's main successes.
- **Takeaway.** The paper provides an early equilibrium asset-pricing framework in which prospect-theory preferences over financial gains and losses can rationalize major aggregate stock market puzzles.

2 Incremental contribution of the paper

2.1 Contribution relative to consumption-based asset pricing

- Earlier consumption-based models try to explain stock returns using utility over consumption alone, often with power utility, Epstein-Zin recursive utility, or habit formation.
- Barberis, Huang, and Santos add a new utility source: direct utility from financial wealth changes.
- This helps explain why stock returns can be volatile and weakly correlated with consumption growth even when consumption itself is smooth.

Interpretation: The incremental step is to stop forcing all stock market movements to be justified by consumption risk. The model gives stock price movements their own psychological payoff channel.

2.2 Contribution relative to prospect theory and behavioral finance

- Prospect theory had been influential in psychology and behavioral decision theory, but less often embedded in a full dynamic equilibrium asset pricing model.
- The paper translates loss aversion and reference dependence into a representative-agent equilibrium with stock prices and risk-free rates.

- It shows how laboratory evidence on losses and prior gains can generate time-varying aggregate risk aversion.

Interpretation: The paper does not merely use prospect theory as a narrative explanation. It builds a model whose equilibrium price-dividend ratios and return moments can be compared with the data.

2.3 Contribution relative to loss-aversion models without dynamics

- A static loss-aversion model can help explain why stocks require a high average premium.
- But static loss aversion cannot naturally generate the strong variation in expected returns and volatility found in the data.
- The paper adds a dynamic state variable z_t that tracks prior gains and losses and thereby changes the degree of loss aversion over time.

Interpretation: The important contribution is the link between past returns and current risk tolerance. This produces return predictability and volatile discount rates.

2.4 Contribution relative to habit and time-varying risk-aversion models

- Habit models also generate time-varying risk aversion, but the state variable is typically consumption relative to habit.
- In this paper, the state variable is past investment performance.
- This makes risk aversion directly responsive to stock market gains and losses rather than only to consumption history.

Interpretation: The model offers a financial-wealth-based alternative to consumption-habit explanations of time-varying expected returns.

2.5 Broader contribution

- The paper helps connect asset pricing to behavioral evidence on the house-money effect and break-even effects.
- It shows that the equity premium, excess volatility, return predictability, and low consumption-return correlation can be studied in one unified behavioral framework.
- It motivates later research on reference-dependent preferences, realized gains and losses, and investor sentiment in asset prices.

3 Model environment and preferences

3.1 Assets and endowments

- There is one risk-free asset and one risky asset.
- In the first economy, the risky asset is a claim to aggregate consumption. This is the closest analogue of the standard consumption-based model.
- In the second economy, dividends are separated from consumption and follow their own process. This is more realistic because actual stock dividends are much more volatile than aggregate consumption.

- The risky asset pays dividends D_t , has price P_t , and gross return

$$R_{t+1} = \frac{P_{t+1} + D_{t+1}}{P_t}. \quad (1)$$

Interpretation: The first economy shows the basic mechanism but is too restrictive. The second economy is where the model can better match stock return moments.

3.2 Consumption utility

The investor receives standard power utility from consumption:

$$u(C_t) = \frac{C_t^{1-\gamma}}{1-\gamma}, \quad (2)$$

where γ is relative risk aversion over consumption.

Interpretation: The paper deliberately keeps consumption utility simple so that the new behavioral channel is easy to isolate.

3.3 Utility from financial gains and losses

The investor also obtains direct utility from gains and losses in financial wealth. Let X_{t+1} denote the financial gain or loss from holding the risky asset from t to $t+1$. A basic loss-aversion specification is

$$v(X_{t+1}) = \begin{cases} X_{t+1}, & X_{t+1} \geq 0, \\ \lambda X_{t+1}, & X_{t+1} < 0, \end{cases} \quad (3)$$

where $\lambda > 1$.

Interpretation: The kink at zero captures loss aversion. Losses are weighted more heavily than gains.

3.4 Prior gains and losses

The paper adds a state variable z_t measuring prior gains and losses.

- $z_t = 1$: the investor has neither prior gains nor prior losses.
- $z_t < 1$: the investor has prior gains. Losses are less painful because they are cushioned by earlier gains.
- $z_t > 1$: the investor has prior losses. Additional losses are more painful because the investor has already been burned.

The gain-loss utility is written as

$$v(X_{t+1}, S_t, z_t), \quad (4)$$

where S_t is the value invested in the stock market.

Interpretation: This state dependence is the source of time-varying risk aversion in the model.

3.5 Reference level

The investor evaluates gains and losses relative to a reference return, usually the risk-free return. A stock return below the reference point is treated as a loss, while a return above the reference point is treated as a gain.

Interpretation: This is important because a stock market decline is not the only possible loss. Even a stock return below the safe alternative can be coded as a loss.

4 Equilibrium pricing

4.1 Price-dividend ratio as a function of the state

The equilibrium price-dividend ratio is a function of the state variable:

$$\frac{P_t}{D_t} = f(z_t). \quad (5)$$

The return can then be written as

$$R_{t+1} = \frac{1 + f(z_{t+1})}{f(z_t)} \frac{D_{t+1}}{D_t}. \quad (6)$$

Interpretation: When prior gains are high, z_t is low, risk aversion is low, and the equilibrium price-dividend ratio is high. When prior losses are high, z_t is high and the price-dividend ratio is low.

4.2 State dynamics

The state variable z_t evolves with realized stock market performance. A good stock return lowers future loss aversion by creating prior gains; a poor stock return raises future loss aversion by creating prior losses.

In simplified form,

$$z_{t+1} = h(z_t, R_{t+1} - R_f), \quad (7)$$

where R_f is the risk-free return.

Interpretation: The equilibrium is recursive: prices determine returns, returns update z_t , and z_t determines future prices.

4.3 Economic intuition for predictability

- A high price-dividend ratio means the market recently performed well.
- Good prior performance reduces loss aversion.
- Lower loss aversion raises prices and lowers expected future returns.
- Therefore high price-dividend ratios forecast low future returns.

Interpretation: This is the mechanism behind the return predictability results in Tables V and VI.

5 Economy I: consumption equals dividends

5.1 Purpose of Economy I

Economy I is a benchmark in which the risky asset is a claim to consumption. This allows the authors to compare their model directly with the standard consumption-based model.

- Consumption growth has mean g_C and volatility σ_C .
- The investor has power utility from consumption and prospect-theory utility from financial gains and losses.
- The key behavioral parameters are λ , k , b_0 , and η .

Interpretation: Economy I illustrates the core idea but cannot fully match stock market volatility because consumption growth is too smooth.

5.2 Findings in Economy I

- Prior gains lower loss aversion, so price-dividend ratios are high when z_t is low.
- Prior losses raise loss aversion, so price-dividend ratios are low when z_t is high.
- The model can raise the equity premium, but return volatility remains too low in reasonable parameter ranges.

Interpretation: Economy I shows that dynamic loss aversion works in the right direction, but tying dividends to consumption makes returns too smooth.

6 Economy II: separating dividends and consumption

6.1 Why dividends and consumption must be separated

In the data, stock dividends are much more volatile than aggregate consumption. If stocks are modeled as claims to consumption, the model mechanically underpredicts return volatility. Economy II therefore allows dividends to follow a separate process:

$$\Delta d_t = g_D + \sigma_D \varepsilon_t^D, \quad (8)$$

while consumption follows

$$\Delta c_t = g_C + \sigma_C \varepsilon_t^C. \quad (9)$$

The two shocks can be correlated.

Interpretation: Separating consumption and dividends is crucial. It allows stock cash flows to be volatile without making consumption implausibly volatile.

6.2 Findings in Economy II

- The model generates a much higher equity premium.
- Stock return volatility rises substantially.
- The correlation between stock returns and consumption growth becomes low, closer to the data.
- Price-dividend ratios predict long-horizon returns.
- Price-dividend ratios are persistent because the state variable z_t is persistent.

Interpretation: Economy II is the main quantitative success of the paper.

7 Sensitivity analysis and the role of prior outcomes

7.1 Parameter k

The parameter k governs how rapidly loss aversion increases after prior losses. Larger k makes investors much more sensitive after poor past performance. Table VII shows that raising k increases the equity premium and return volatility.

7.2 Loss aversion λ

The parameter λ controls the basic kink in the value function. Table VIII shows that higher λ raises the equity premium because losses become more painful.

7.3 Persistence η

The parameter η controls how long prior outcomes affect risk aversion. Table IX shows that more persistence raises return volatility and price-dividend autocorrelation.

7.4 Evaluation period and reference level

Tables X and XI show that results depend on how often gains and losses are evaluated and on the reference return. Shorter evaluation periods and higher reference levels make losses more salient and raise required premia.

Interpretation: The sensitivity tables show that the model's performance depends on psychologically meaningful parameters: loss aversion, memory, evaluation horizon, and reference point.

7.5 Why prior outcomes are essential

The final section removes the effect of prior outcomes and lets loss aversion be constant over time. The model can still generate some equity premium, but it fails to generate enough volatility.

Interpretation: This isolates the central contribution: loss aversion alone is not enough. The interaction between loss aversion and prior outcomes is what helps explain price dynamics.

8 Tables and figures

8.1 Figures I and II: prospect theory value functions

Read:

- Figure I shows how the value function changes with prior gains or losses.
- Prior gains flatten the loss region because losses are cushioned by past gains.
- Prior losses steepen the loss region because the investor is already in a painful loss state.
- Figure II shows the Kahneman-Tversky value function: concave over gains, convex over losses, and kinked at the reference point.

Economic message: These figures provide the psychological foundation for the model. Asset prices move because the investor's tolerance for risk depends on prior gains and losses.

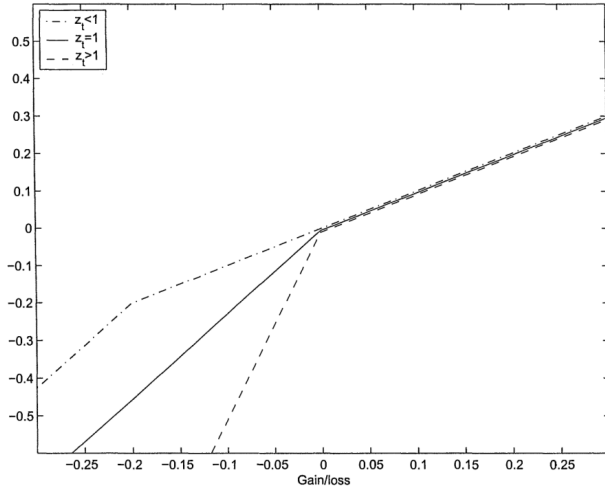


FIGURE I

Utility of Gains and Losses

The dash-dot line represents the case where the investor has prior gains, the dashed line the case of prior losses, and the solid line the case where he has

Figure I: utility of gains and losses

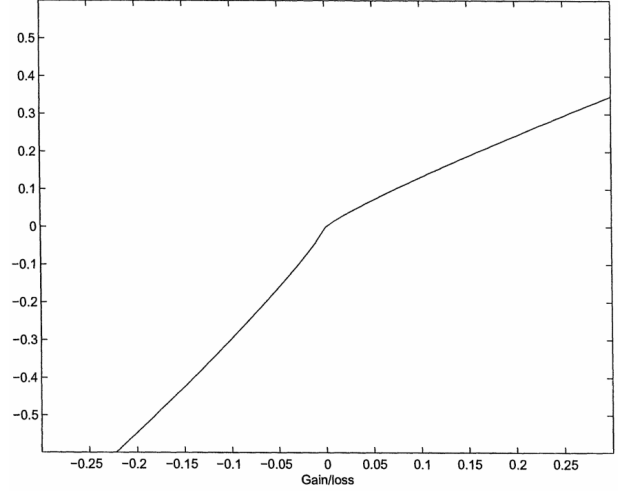


FIGURE II

Kahneman-Tversky Value Function

Figure II: Kahneman-Tversky value function

8.2 Table I and Figure III: Economy I benchmark

Read:

- Table I lists the parameters for Economy I, including g_C , σ_C , γ , ρ , λ , k , b_0 , and η .
- Figure III plots price-dividend ratios as functions of the state variable z_t .
- Price-dividend ratios are high after prior gains and low after prior losses.

Economic message: The benchmark economy shows the central state-price mechanism: past returns move risk aversion, and risk aversion moves the price-dividend ratio.

TABLE I PARAMETER VALUES FOR ECONOMY I	
Parameter	
g_C	1.84%
σ_C	3.79%
γ	1.0
ρ	0.98
λ	2.25
k	(range)
b_0	(range)
η	0.9

Table I: parameters for Economy I

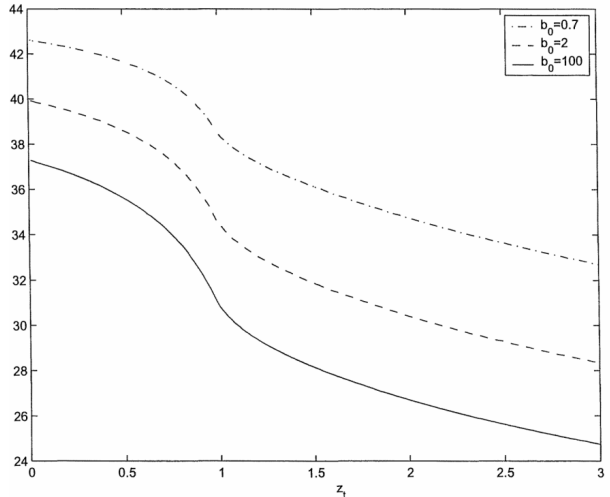


FIGURE III

Price-Dividend Ratios in Economy I

The price-dividend ratios are plotted against z_t , which measures prior gains and losses: a low z_t indicates prior gains. The parameter b_0 controls how much the investor cares about financial wealth fluctuations. We fix the parameter k at 3, implying average loss aversion close to 2.25 in all cases.

Figure III: price-dividend ratios in Economy I

8.3 Figure IV and Table II: state distribution and return moments in Economy I

Read:

- Figure IV shows the equilibrium distribution of z_t .
- The distribution is centered near one but has right-tail states corresponding to prior losses.
- Table II reports asset return moments in Economy I for different values of b_0 and k .
- The model can raise the equity premium but struggles to generate enough return volatility.

Economic message: Economy I has the right qualitative mechanism but too little cash-flow volatility, so stock return volatility is too low.

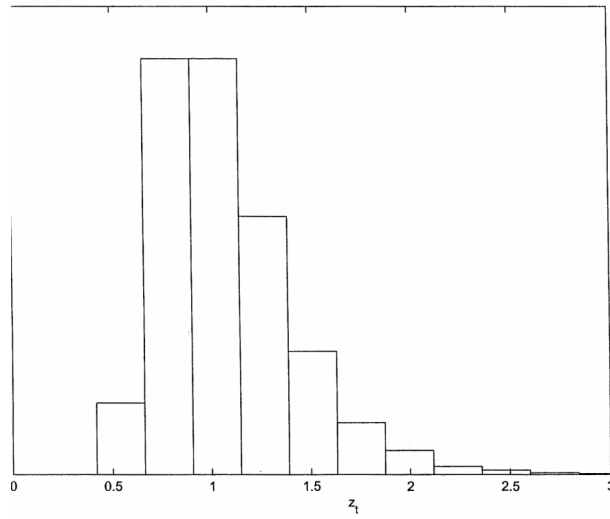


FIGURE IV
Distribution of the State Variable z_t

The distribution is based on Economy I with $b_0 = 2$ and $k = 3$.

Figure IV: distribution of z_t

TABLE II
ASSET RETURNS IN ECONOMY I

	$b_0 = 0$	$b_0 = 0.7$	$b_0 = 2$	$b_0 = 100$	Empirical
		$k = 3$	$k = 3$	$k = 3$	value
Log risk-free rate	3.79	3.79	3.79	3.79	0.58
Log excess stock return					
Mean	0.07	0.63	0.88	1.26	6.03
Standard deviation	3.79	4.77	5.17	5.62	20.02
Sharpe ratio	0.02	0.13	0.17	0.22	0.3
Average loss aversion		2.25	2.25	2.25	
		$b_0 = 0.7$	$b_0 = 2$	$b_0 = 100$	
		$k = 150$	$k = 100$	$k = 50$	
Log risk-free rate		3.79	3.79	3.79	
Log excess stock return					
Mean		3.50	3.66	3.28	
Standard deviation		10.43	10.22	9.35	
Sharpe ratio		0.34	0.36	0.35	
Average loss aversion		10.7	7.5	4.4	

Moments of asset returns are expressed as annual percentages. Empirical values are based on Treasury Bill and NYSE data from 1926–1995. The parameter b_0 controls how much the investor cares about financial wealth fluctuations, while k controls the increase in loss aversion after a prior loss.

Table II: asset returns in Economy I

8.4 Table III and Figure V: Economy II

Read:

- Table III lists parameters for Economy II and compares them with Economy I.
- Economy II introduces separate dividend growth, with mean g_D , volatility σ_D , and correlation ω with consumption growth.
- Figure V shows price-dividend ratios in Economy II.

Economic message: Separating dividends from consumption makes the model more realistic and improves the quantitative fit.

TABLE III
PARAMETER VALUES FOR ECONOMY II

Parameter	Economy II	Economy I
g_C	1.84%	1.84%
g_D	1.84%	—
σ_C	3.79%	3.79%
σ_D	12.0%	—
ω	0.15	—
γ	1.0	1.0
ρ	0.98	0.98
λ	2.25	2.25
k	(range)	(range)
b_0	(range)	(range)
η	0.9	0.9

The parameter values used for Economy I and shown in Table I are repeated here for ease of comparison.

Table III: parameters for Economy II

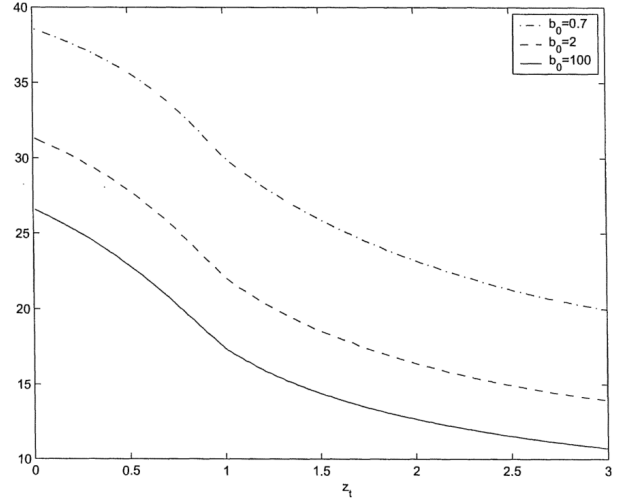


FIGURE V
Price-Dividend Ratios in Economy II

The price-dividend ratios are plotted against z_t , which measures prior gains and losses: a low z_t indicates prior gains. The parameter b_0 controls how much the investor cares about financial wealth fluctuations. We fix the parameter k at 3, giving average loss aversion close to 2.25 in all cases.

Figure V: price-dividend ratios in Economy II

8.5 Table IV and Figure VI: asset prices and conditional moments

Read:

- Table IV reports the main asset-pricing moments for Economy II.
- With dynamic loss aversion, the model generates a high equity premium, higher stock volatility, low correlation with consumption growth, and variable price-dividend ratios.
- Figure VI shows conditional expected returns and conditional volatility as functions of z_t .
- Expected returns rise after prior losses because investors require more compensation for risk.

Economic message: This is the model's main success: it explains high and time-varying expected returns through state-dependent risk aversion.

TABLE IV
ASSET PRICES AND RETURNS IN ECONOMY II

	$b_0 = 0$ $k = 3$	$b_0 = 0.7$ $k = 3$	$b_0 = 2$ $k = 3$	$b_0 = 100$ $k = 3$	Empirical value
Log risk-free rate	3.79	3.79	3.79	3.79	0.58
Log excess stock return					
Mean	-0.65	1.3	2.62	3.68	6.03
Standard deviation	12.0	17.39	20.87	20.47	20.02
Sharpe ratio	-0.05	0.07	0.13	0.18	0.3
Correlation w/consumption growth	0.15	0.15	0.15	0.15	0.1
Price-dividend ratio					
Mean	76.6	29.8	22.1	17.5	25.5
Standard deviation	0	2.9	2.7	2.4	7.1
Average loss aversion		2.25	2.25	2.25	

	$b_0 = 0.7$ $k = 20$	$b_0 = 2$ $k = 10$	$b_0 = 100$ $k = 8$
Log risk-free rate	3.79	3.79	3.79
Log excess stock return			
Mean	5.17	5.02	5.88
Standard deviation	25.85	23.84	24.04
Sharpe ratio	0.2	0.21	0.24
Correlation w/consumption growth	0.15	0.15	0.15
Price-dividend ratio			
Mean	14.0	14.6	12.7
Standard deviation	2.6	2.5	2.2
Average loss aversion	5.8	3.5	3.2

Moments of asset returns are expressed as annual percentages. Empirical values are based on Treasury Bill and NYSE data from 1926–1995. The parameter b_0 controls how much the investor cares about financial

Table IV: asset prices and returns in Economy II

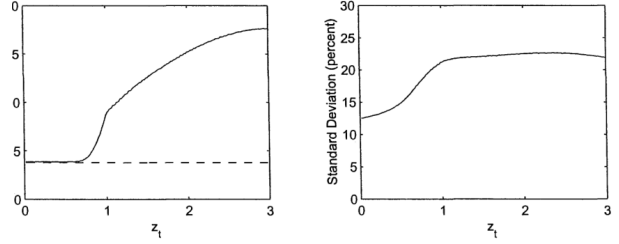


FIGURE VI
Conditional Moments of Stock Returns

The conditional stock return mean and volatility are plotted against z_t , which assures prior gains and losses: a low z_t indicates prior gains. The results are for Economy II with $b_0 = 2$ and $k = 3$. The dashed line in the left-hand panel

Figure VI: conditional moments of stock returns

8.6 Tables V and VI: return dynamics and predictability

Read:

- Table V reports autocorrelations of returns and price-dividend ratios.
- The model generates negative return autocorrelation and persistent price-dividend ratios.
- Table VI reports predictability regressions of cumulative returns on lagged dividend-price ratios.
- Predictability rises with the return horizon, resembling the empirical evidence.

Economic message: The model links high prices to low future returns because high prices follow prior gains and low loss aversion.

TABLE V
AUTOCORRELATIONS OF LOG RETURNS AND PRICE-DIVIDEND RATIOS IN ECONOMY II

	$b_0 = 2$ $k = 3$	$b_0 = 2$ $k = 10$	Empirical value
$\text{corr}(r_t, r_{t-j})$			
$j = 1$	-0.07	-0.12	0.07
$j = 2$	-0.05	-0.09	-0.17
$j = 3$	-0.04	-0.06	-0.05
$j = 4$	-0.04	-0.04	-0.11
$j = 5$	-0.02	-0.03	-0.04
$\text{corr}((P/D)_t, (P/D)_{t-j})$			
$j = 1$	0.81	0.72	0.70
$j = 2$	0.66	0.52	0.50
$j = 3$	0.53	0.38	0.45
$j = 4$	0.43	0.28	0.43
$j = 5$	0.35	0.20	0.40

Empirical values are based on NYSE data from 1926–1995.

Table V: autocorrelations

TABLE VI
RETURN PREDICTABILITY REGRESSIONS IN ECONOMY II

	$b_0 = 2$ $k = 3$	$b_0 = 2$ $k = 10$	Empirical value
$\beta_j, R^2(j)$			
β_1	4.6	4.4	4.2
β_2	8.3	7.5	8.7
β_3	11.6	9.7	12.1
β_4	13.7	11.5	15.9
$R^2(1)$	2%	6%	7%
$R^2(2)$	4%	10%	16%
$R^2(3)$	5%	12%	22%
$R^2(4)$	6%	14%	30%

Coefficients and R^2 in regressions of j -year cumulative log returns on the lagged dividend-price ratio, $r_{t+1} + r_{t+2} + \dots + r_{t+j} = \alpha_j + \beta_j(D_t/P_t) + \epsilon_{j,t}$. Empirical values are based on annual NYSE data from 1926–1995.

Table VI: return predictability regressions

8.7 Tables VII and VIII: sensitivity to k and λ

Read:

- Table VII varies k , which controls how loss aversion rises after prior losses.

- Higher k raises the equity premium and return volatility.
- Table VIII varies λ , the basic loss-aversion parameter.
- Higher λ raises the equity premium by making losses more painful.

Economic message: The most important behavioral parameters govern the marginal pain of losses and how that pain varies after bad performance.

TABLE VII
SENSITIVITY OF ASSET RETURNS TO k

	$k = 3$	$k = 5$	$k = 10$	Empirical value
Log excess stock return				
Mean	2.62	3.15	5.02	6.03
Standard deviation	20.87	20.93	23.84	20.02
Sharpe ratio	0.13	0.15	0.21	0.3
Average loss aversion	2.25	2.6	3.5	

The parameter k controls how much loss aversion increases after a prior loss. Moments of asset returns are expressed as annual percentages. The results are for Economy II with $b_0 = 2$; other parameters are fixed at the values in Table III.

Table VII: sensitivity to k

TABLE VIII
SENSITIVITY OF ASSET RETURNS TO λ

	$\lambda = 1.5$	$\lambda = 2.25$	$\lambda = 3$	Empirical value
Log excess stock return				
Mean	3.8	5.02	5.6	6.03
Standard deviation	25.68	23.84	23.21	20.02
Sharpe ratio	0.15	0.21	0.24	0.3
Average loss aversion	3.2	3.5	3.9	

The parameter λ controls the investor's loss aversion. Moments of asset returns are expressed as annual

Table VIII: sensitivity to λ

8.8 Tables IX and X: persistence and evaluation period

Read:

- Table IX varies η , which determines how long prior outcomes affect risk aversion.
- Higher persistence increases return volatility and price-dividend persistence.
- Table X varies the evaluation period over which investors assess gains and losses.
- Shorter evaluation periods generally make the stock market feel riskier.

Economic message: The model's predictions depend not only on loss aversion but also on the mental accounting horizon.

TABLE IX
SENSITIVITY OF ASSET RETURNS TO η

	$\eta = 1$	$\eta = 0.9$	$\eta = 0.8$	Empirical value
Log excess stock return				
Mean	7.68	5.02	3.91	6.03
Standard deviation	34.54	23.84	19.12	20.02
Sharpe ratio	0.22	0.21	0.2	0.3
Autocorrelation of price-dividend ratio	0.81	0.72	0.65	0.7
Average loss aversion	4.5	3.5	3.0	

The parameter η governs how long-lasting the effects of prior gains and losses are. Moments of asset returns are expressed as annual percentages. The results are for Economy II with $b_0 = 2$ and $k = 10$; other parameters are fixed at the values in Table III.

Table IX: sensitivity to η

TABLE X
SENSITIVITY OF ASSET RETURNS TO THE EVALUATION PERIOD

	6 months	1 year	2 years	Empirical value
Log excess stock return				
Mean	7.63	5.02	2.85	6.03
Standard deviation	27.78	23.84	20.15	20.02
Sharpe ratio	0.27	0.21	0.14	0.3
Average loss aversion	3.5	3.5	3.6	

The evaluation period is the length of time over which the investor measures his gains and losses. Moments of asset returns are expressed as annual percentages. The results are for Economy II with $b_0 = 2$ and $k = 10$; other parameters are fixed at the values in Table III.

Table X: evaluation period

8.9 Tables XI and XII: reference level and no-prior-outcome model

Read:

- Table XI varies the reference level used to classify gains and losses.
- A lower reference level makes losses less frequent and reduces the equity premium.
- Table XII lists parameters for the model in which prior outcomes have no effect.

Economic message: The reference point matters. More important, removing prior-outcome effects lets the paper isolate how much of the result comes from dynamic loss aversion.

TABLE XI
SENSITIVITY OF ASSET RETURNS TO THE REFERENCE LEVEL

	0%	-1%	-2%	Empirical value
Log excess stock return				
Mean	5.02	4.11	3.43	6.03
Standard deviation	23.84	24.25	25.20	20.02
Sharpe ratio	0.21	0.17	0.14	0.3
Average loss aversion	3.5	3.5	3.7	

A "0%" reference level means that the investor compares the stock return with the risk-free rate when measuring gains and losses. "-1%" means a reference level one percent lower than the risk-free rate. Moments of asset returns are expressed as annual percentages. The results are for Economy II with $b_0 = 2$ and $k = 10$; other parameters are fixed at the values in Table III.

Table XI: reference level

TABLE XII
PARAMETER VALUES FOR A MODEL WHERE PRIOR OUTCOMES HAVE NO EFFECT

Parameter	Section VI	Subsection V.D
g_C	1.84%	1.84%
g_D	1.84%	1.84%
σ_C	3.79%	3.79%
σ_D	12.0%	12.0%
γ	1.0	1.0
ρ	0.98	0.98
λ	2.25	2.25
k	—	(range)
b_0	(range)	(range)
η	—	0.9

Parameter values used for Economy II in subsection V.D are presented alongside for comparison.

Table XII: parameters without prior-outcome effects

8.10 Figure VII and Table XIII: why prior outcomes matter

Read:

- Figure VII plots price-dividend ratios and expected returns when prior outcomes have no effect.
- Table XIII reports asset return moments in the same model.
- The model can generate some equity premium but fails to generate enough stock return volatility.

Economic message: This final comparison shows that the prior-outcome channel is essential. Static loss aversion alone cannot explain the joint behavior of returns, volatility, and predictability.

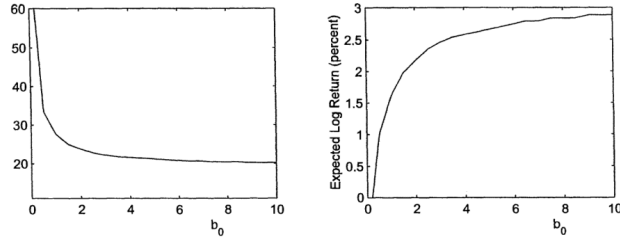


FIGURE VII
Stock Prices and Returns When Prior Outcomes Have No Effect
The price-dividend ratio and the equity premium are plotted against b_0 , which
Figure VII: stock prices and returns without prior-outcome effects

TABLE XIII
ASSET RETURNS WHEN PRIOR OUTCOMES HAVE NO EFFECT

	$b_0 = 0$	$b_0 = 0.7$	$b_0 = 2$	$b_0 = 100$	Empirical value
Log risk-free rate	3.79	3.79	3.79	3.79	0.58
Log excess stock return					
Mean	-0.65	1.30	2.17	2.88	6.03
Standard deviation	12.0	12.0	12.0	12.0	20.02
Sharpe ratio	-0.05	0.11	0.18	0.24	0.3

Moments of asset returns are expressed as annual percentages. Empirical values are based on Treasury Bill and NYSE data from 1926–1995. The parameter b_0 controls how much the investor cares about financial wealth fluctuations.

Table XIII: asset returns without prior-outcome effects

9 Short summary

- The paper adds prospect-theory utility over financial gains and losses to an asset-pricing model.
- Investors are loss averse, and loss aversion depends on prior investment outcomes.
- Prior gains reduce future loss aversion; prior losses increase future loss aversion.
- Time-varying loss aversion generates time-varying expected returns.
- High price-dividend ratios predict low future returns because high prices follow prior gains and low risk aversion.
- Separating dividends from consumption is necessary for matching high stock return volatility and low consumption-return correlation.

- The model helps explain the equity premium, excess volatility, predictability, and low correlation of returns with consumption growth.
- The model without prior outcomes performs much worse, showing that dynamic reference dependence is central.

10 Conclusion

10.1 Core conclusion

Barberis, Huang, and Santos show that prospect-theory preferences can help explain major aggregate stock market facts. Investors care directly about stock market gains and losses, not only about consumption. Because their sensitivity to losses depends on prior investment performance, aggregate risk aversion changes over time.

10.2 Economic intuition

When the market rises, investors feel cushioned by prior gains and become more willing to bear risk. Prices rise and expected returns fall. When the market falls, investors become more loss averse. Prices fall and expected returns rise. These shifts in risk aversion create excess volatility and return predictability.

10.3 Takeaway

The paper's key lesson is that asset prices may reflect reference-dependent utility from financial wealth fluctuations. Consumption risk alone is not enough to explain the stock market; investors' reactions to gains, losses, and prior outcomes can move discount rates and thereby move prices.